Panasonic

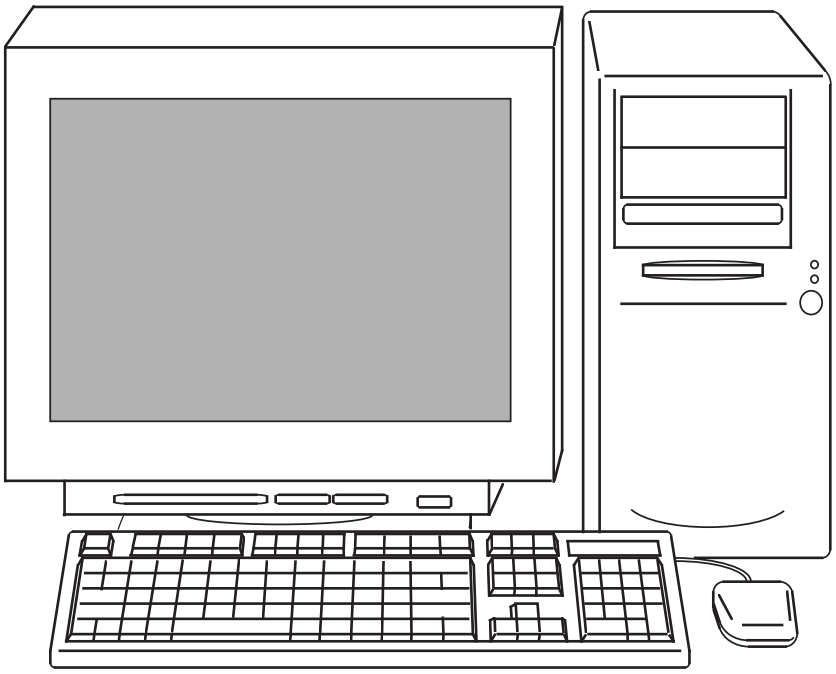
Operating Instructions

Basic Operations

MES for Electronics Assembly

PanaCIM-EE

Model No. **NM-EJS3B**



Thank you very much for purchasing “Panasonic Product”.

- Read this operating instructions carefully to use the product safely and correctly.
- Read “Safety Instructions” before you use the product.
- Store this instructions in a safe place.

N7201A441E10 (V08.08)
201801

Original Instructions

For using this system:
This system is designed with a purpose of visualization of work productivity, more efficiency and quality warranty in electronic component mounting process. Use this system solely for the purpose.

Do not use this system for other purposes.

Operating machines:
Any person who operates the machine should receive a special training.

Any operator who has not received the training is not allowed to operate.

For training, contact Panasonic Smart Factory Solutions Co., Ltd., Sales company or Sales agent.

⚠ Warning

Do not operate the machine without receiving special training



Risk of injury due to improper operation.

Prohibited

●About training, conduct in your company or contact us.

For machine training in Europe, you may contact Panasonic Industry Europe:

Panasonic Industry Europe GmbH FA Division
Robert-Koch-Strasse 100, 85521 Ottobrunn Germany
Phone number: 0049-(0)89-453541192

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About the Instruction

■ Operating instruction contents

The following operating instructions are included in the DVD-ROM attached to this machine. The instructions are divided according to work or function. Use one that is appropriate for the purchased module (application) or your purpose.

∗ BM series machine is included in the explanation of this manual, but its functions are not supported.

Type	Description
<General>	
Introduction	Overview of machine and basic setting/operation/analysis/utility.
Basic Operation (PanaCIM Web Application)	PanaCIM Web application's basic operation.
<Modules>	
Production Analysis	Collect and analyze machine's running information and display its statistics on real time basis to support productivity improvements (recognition/pickup error rate improvement, machine trouble stop time reduction) for mounting line.
Traceability	Combine PCB information read by a barcode reader on machine or conveyor and production record or material/part information of machine to manage history of PCBs for quality control and market support.
Material Verification	Verify material/part information and recipes while setting up material/parts in production setup/model change/splicing/part exhaust to prevent wrong material/part set up.
Product Changeover and Control	Auto download recipe and adjust lane width based on PCB ID for quicker model change.
Production Monitoring	Retrieve production information and operation information on real time basis and display line status on the monitor and PDA to provide operators with instructions for events such as error or part exhaust. This is to reduce mount line stop time or to prepare in advance for better productivity.
Enterprise Link	A function to output the data collected from machines during production (material verification data, trace data, operation performance data) in the standard format so external applications can reuse the data. Part information can be loaded also in the standard format.

Type	Description
Material Control	A function to provide the location and the number of parts or feeders stocked in the production or preparation area. MSD function controls MSD parts' expired date.
Component Supply Navigator	The Component Supply Navigator optimizes the supply sequence to provide a shortest route of an operator without component shortage. It can prevent short time breakdown in the production line occurred due to component shortage and contribute to manpower saving.
Maintenance	For the purpose of reducing time spent on maintenance planning / control operations, asset (machines, heads and feeders) profiles can be controlled on the floor, or the maintenance history of each asset can be recorded.

License CD names for modules are as follows.

Module Name	License CD Name
Production Analysis	Production Analysis
Traceability	Traceability
Material Verification	Material Verification
Product Changeover and Control	Product Changeover and Control
Production Monitoring	Production Monitoring & Dispatch
Enterprise Link	Enterprise Link
Material Control Base	Material Control Base
Material Control Allocation	Material Control Allocation
Material Control MSD	Material Control MSD
Component Supply Navigator	Component Supply Navigator
Maintenance	Maintenance

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1. OVERVIEW

This chapter provides the overview of PanaCIM-EE WEB application.

1.1 Overview

PanaCIM Web application is a web-based system that creates running status/traceability/changeover report for mount process line or machines. Factory's running status can be checked from anywhere inside or outside the factory with a computer connected with internet/intranet.



1.1.1 Browser Compatibility

PanaCIM Web application accesses to other systems. Use Internet Explorer 11.0 later.

Note: If you use a button or navigation of the browser to go to another page, the contents edited on the previous page may be lost or application error may be caused. Use move button of the application to move to another page.

1.2 Function Overview

1.2.1 PanaCIM-EE Web Application Top Page



- Top Page Description

Number	Description
1	Logged-in ‘Server Name/User Name’ appears at ‘Welcome’ on the upper right of the screen.
2	At ‘Database’, select a PanaCIM database you want to access. • **Active**: Shows the latest data. • **Historical:** Shows the archive data.
3	Select ‘Logout’ to close the application.
4	Select any one among **Charts and Reports, Alarms and Exceptions, Traceability, Administration** from Navigation menu.
5	Launch menu appears according to the selected navigation menu.

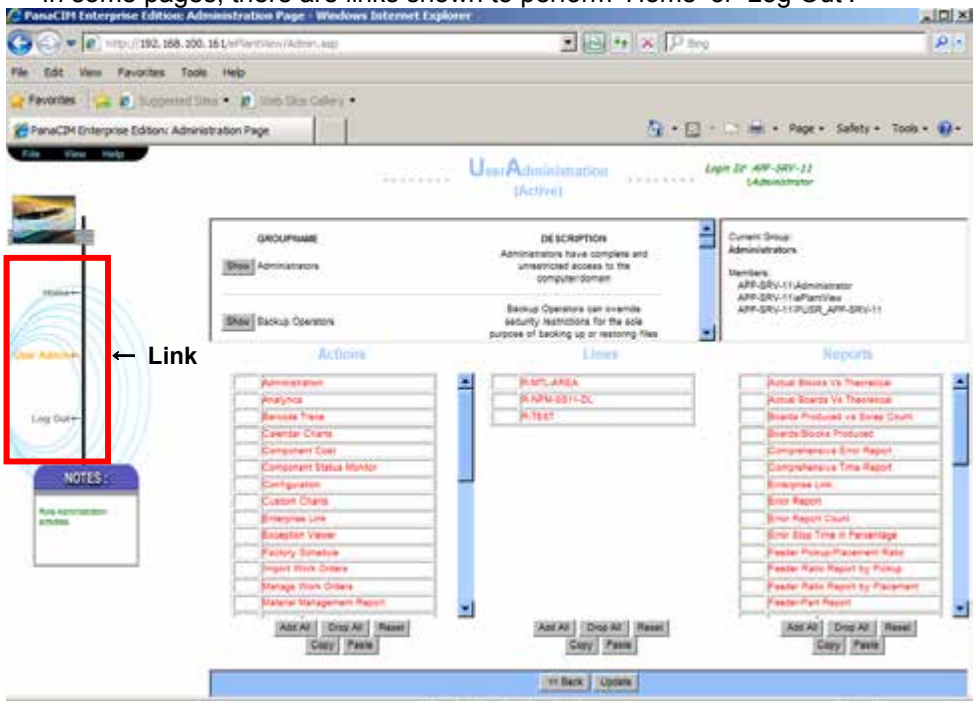
- **List of Navigation and Launch Menu**

Navigation Menu	Launch Menu
Charts and Reports	Reports
	Manage Work Orders
Alarms and Exceptions	Show exception
	Monitoring
Traceability	PCB repair
	Pattern scan
Administration	Work order configuration
	User management
	Task management
	Calendar setting
	Shift setting
	Buffer setting

Note: The menu displayed (selectable) depends on the license you purchased.

1.2.2 Explanation of Link

In some pages, there are links shown to perform ‘Home’ or ‘Log Out’.



■ Link

Link	Description
Page Link (Page Left)	The link to a page accessible. The current page is shown in orange italic.
Home	Close the current page and go back to the top page.
Logout	Close the application and the browser window.

2. SETTINGS

This chapter describes Login setting to PanaCIM Web application.

2.1 Overview

This section explains how to add an account to log into PanaCIM Web application and change password.

PanaCIM application server has the following three Windows groups prior to installation.

- epvAdministrators
- epvSupervisors
- epvOperators

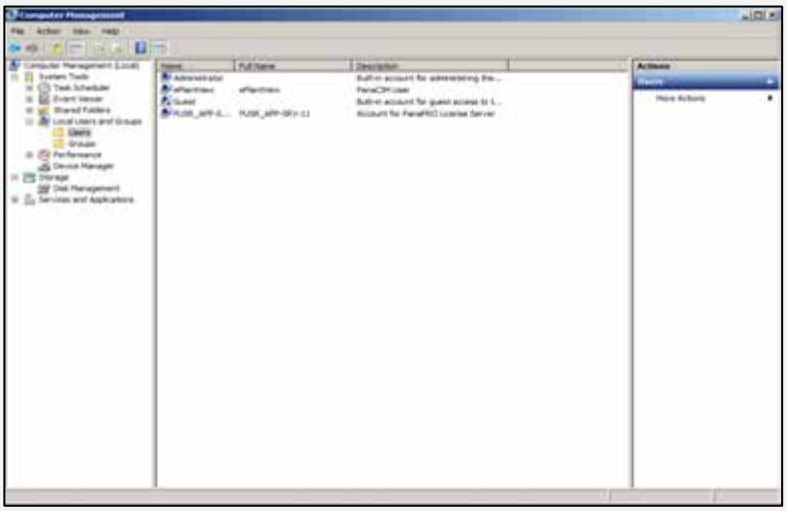
epvAdministrator and Windows account IDs belong to the group are authorized to access all functions and to limit functions. (See “5.2.1 User Administration” for limiting functions.)

2.2 Settings

2.2.1 Add/Delete Account

Add or delete account as follows.

- **Add**

Step	Operation
1	Log into PanaCIM application server (Windows Server) as Administrator.
2	Go to **Control Panel** > **Management Tool** > **Computer Management** and open **Local Users and Groups** > **Users** screen. 
3	Select **Action** > **New User** from menu and enter account information you want to add. **Note:** Password is a required field.
4	Select the added account and open **Member Of** tab from Properties dialog.

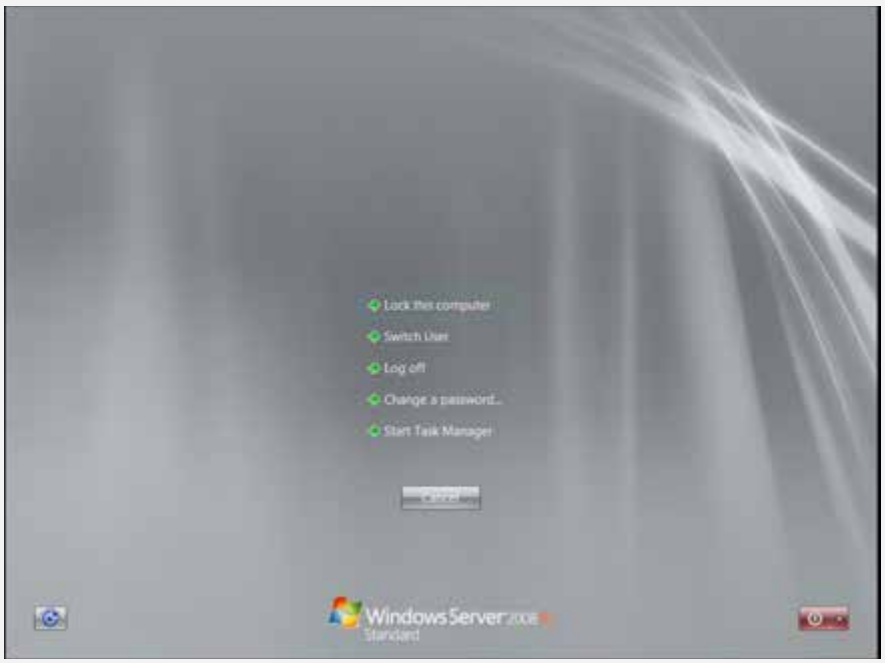
5	Press **Add** button to add a group from epvAdministrators/ epvSupervisors/ epvOperators. 
6	Set password related items on Properties dialog **General** tab and close.

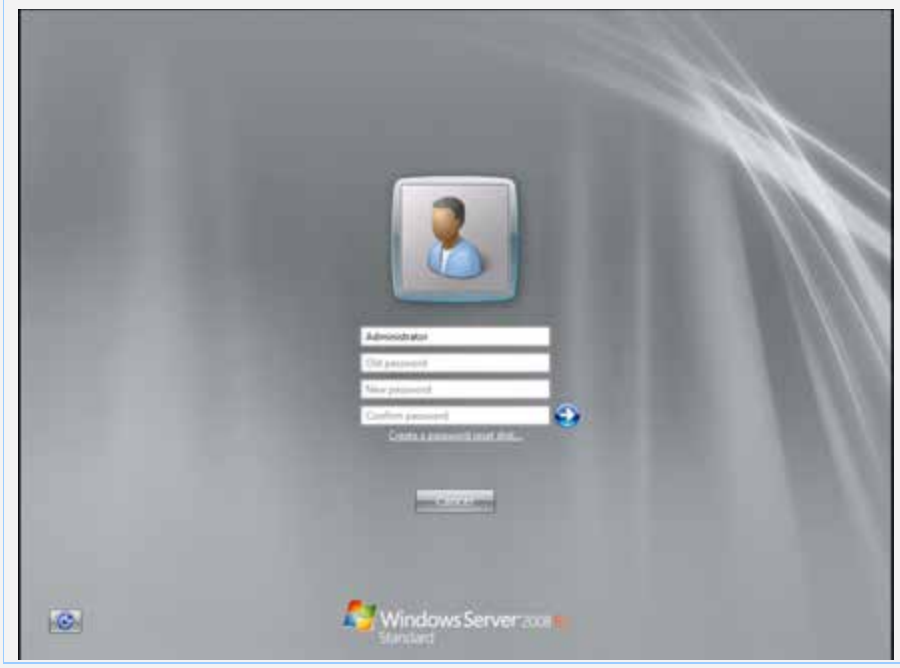
- **Delete**

Step	Operation
1	Go to **Control Panel** > **Management Tool** > **Computer Management** and open **Local Users and Groups** > **Users** screen.
2	Select an account to delete and click **Action** > **Delete** from menu.

2.2.2 Change Password

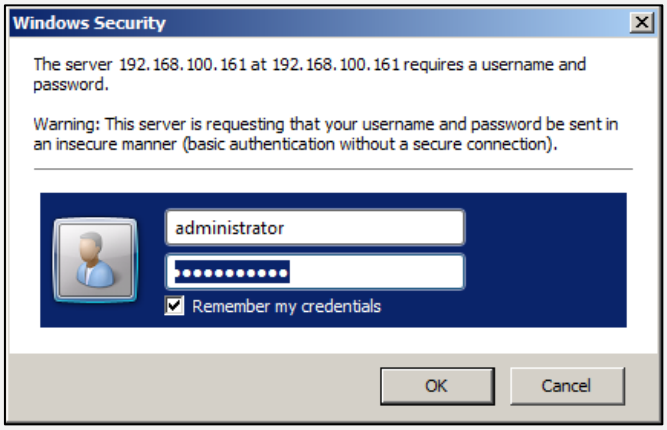
Password can be changed as follows.

Step	Operation
1	Log into PanaCIM application server (Windows Server) with the account you want to change.
2	After login, press Ctrl + Alt + Del key and click **Change a password.** 

Step	Operation
3	Change Password dialog appears. Set a password and close. 

2.2.3 Login

Use PanaCIM application server's Administrator account when you log in PanaCIM Web application for the first time. Default setting allows local administrator to access all lines and reports, and to grant access authorization to all other users.

Step	Operation
1	Open web browser and launch PanaCIM Web application. 1. Accessing from server PC; http://<localhost>/eplantview/ 2. Accessing from PC other than server; http://<PanaCIM App Server Name>/eplantview/
2	Login dialog appears. Enter **User name (Administrator)** and **Password**. 
3	Click **OK** and start PanaCIM Web application.

3. FUNCTION - Charts and Reports

This chapter explains charts and reports of machine's operation performance data.

3.1 Overview

Using the operating performance data of the machine, report type information can be displayed and saved in a report format.



The details of options are explained below.

Launch Menu	Description
Reports	Display and output various reports from the operation performance data.
Manage Work Orders	Display detailed work schedule and allow work orders to be checked, modified and registered. For details, refer to PanaCIM-EE, Material Control.

3.1.1 List of Reports

The table below is the list of available reports.
Note) Even though these modules are selectable, display and output of each report vary depending on your purchased license.

Module/sub-module		Report name
Base		Product Setup Information Report
		Mix Information Report
		Alternate Model Barcodes
		Feeder Allocation Report
		Part Allocation Report
		Work Order History Report
		SP Changeover Report
		Changeover Report
		Supplier Part Number Mapping Report
Material Verification	Base	Component Trace Report
		Current Setup Report
		Work History Report
		Transaction Trace Report
		Production Order Trace Report
		Loaded Feeder Report
		Feeder Routine Check History
		Production Lock History
		Remaining Shelf Time
		LCR Verification History
		Routine Check History
	Pre-setup	Presetup Detail
		Presetup Summary
		Modular Presetup Cart Mount History
		Modular Presetup Cart Current Contents
		Modular Presetup Cart Feeder History
		Presetup Cart Mount Status
		Modular Presetup Cart Check
		Presetup Cart Production Data Check
		Presetup Cart Routine Check
		Presetup Cart ID List
Material Control	Base	Material Quantity Report
		Printer Material Usage Count Report
		Material Adjustment Report
		Material Usage Report
		Material Position History Report
		Feeder Position History Report
		Material Movement Report
		Material Edit Report
		Material Registration Report

Module/sub-module		Report name
		Materials Summary Report
		Part Position Report
		Material Management Report
	MSD	MSD Transaction Report
		Latest MSD Report
		Expired MSD Report
		MSD Part List
	Allocation	Material Allocation Shortage Report
		Material Allocation Reservation Report
		Material Allocation Setup Report
		Material Allocation Production Order History Report
		Allocatable Component Report
Traceability		Panel to Component Traceability
		Barcode Trace Report
		Component to Panel Traceability
		Panel Trace Report
		Panel Hold Trace Report
		Inspection/Repair Trace Report
		Placement Load Measurement Report
Feeder Maintenance		Tagged Feeder Repair Report
		Feeder Repair Count Report
		Feeder Repair History Report
		Feeder Report
Production Analysis		Component Usage Report
		Component Usage Report (incl. costs)
		PCB Production Report
		Work Order Usage Report
		Event Trace Report
		Event Trace Summary Report
		Down Time Report
		Down Time Summary Report
		State Trace Report
		State Trace Summary Report
		Part Supply Patrol Report
Production Monitoring Acknowledgment		PMD Event History Report
		PMD Event Summary Report
		Work in Progress Report
		Upcoming Work Orders Report
		Line Down Time Report
Enterprise Link		Import History Report

3.1.2 Description of Reports

The table below is the descriptions of available reports in Report - [Base].

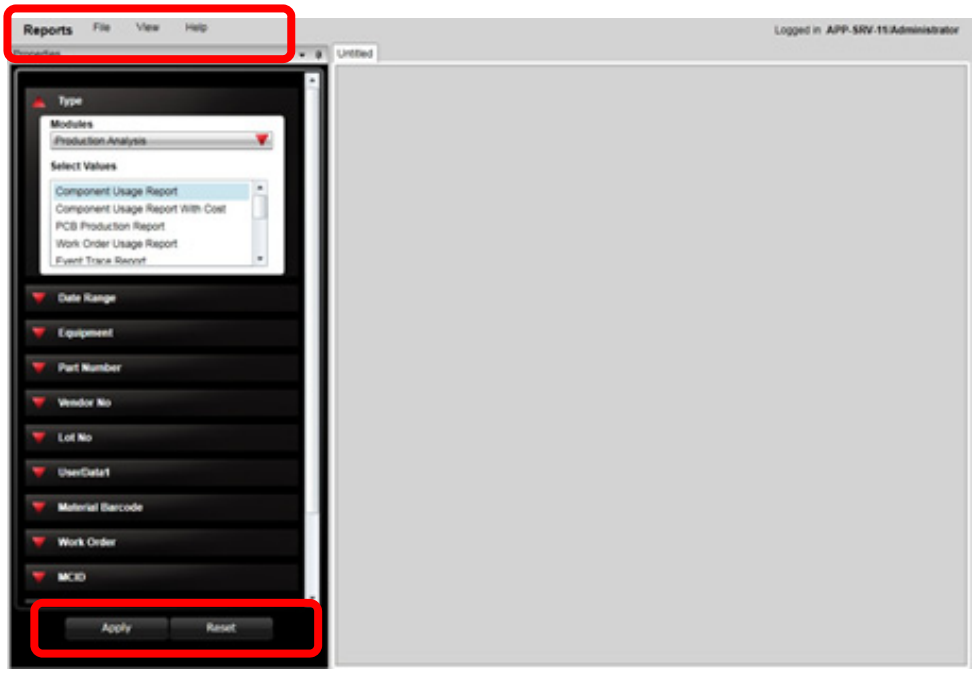
■ Base

Report	Description
Product Setup Information Report	Shows a list of program files for each work cell of the route.
Mix Information Report	Shows a product corresponding to the route selected in the MIX name and machine selection page.
Alternate Model Barcodes	Shows information on substitute product available in production.
Feeder Allocation Report	Shows information on feeders assigned to the product in MIX.
Part Allocation Report	Shows a list of parts assigned to various products in MIX and supplied from the placement machine in the factory.
Work Order History Report	Shows the history data related to the production of all products in work order. Production status can also be checked in this report.
SP Changeover Report	Shows a list of SP changeovers related to machine, product name and corresponding product name.
Changeover Report	Shows all feeders required for producing the selected product on the selected cell during changeover, and their positions on the part supply unit of the machine.
Supplier Part Number Mapping Report	Shows supplier part number for part number.

3.2 Settings

3.2.1 Explanation of File Menu/Button

In the report window, file menus and buttons are displayed as shown below.



The table below shows explanations of file menus and buttons shown in the window.

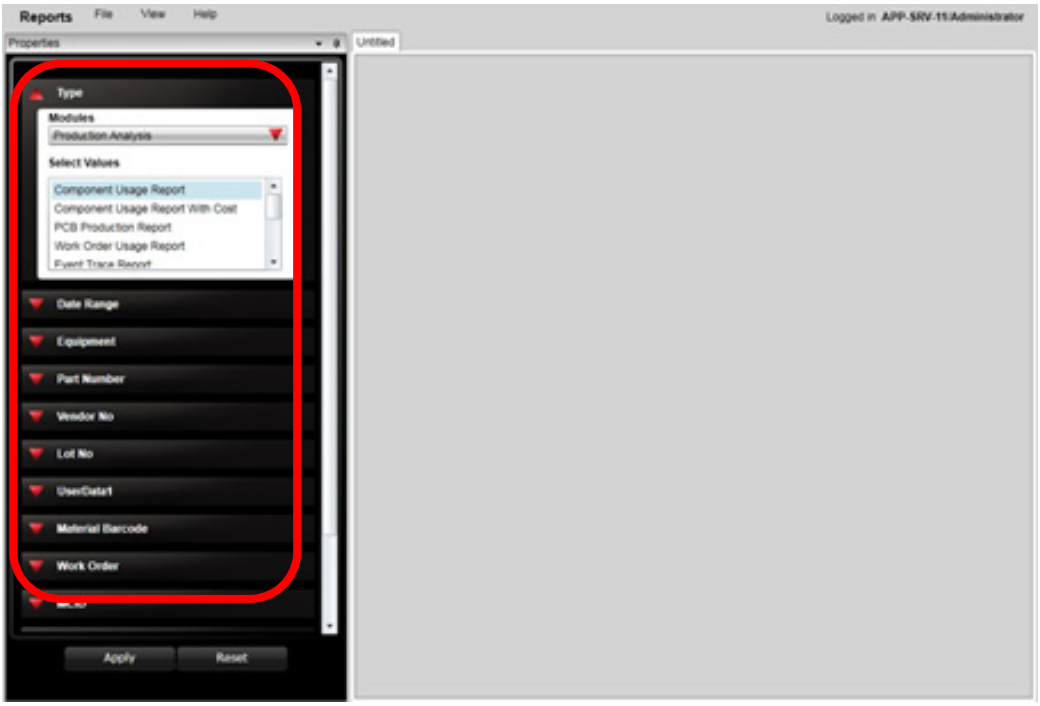
Menu	Sub-menu	Explanation
File	New	Clear information on the current page and display the default report window.
	Open	Filter a saved file and display a report.
	Save	Save a currently displayed filter criteria in a file. When the data is opened from the saved file, it overwrites and saved. If the file is newly saved, it saved as new name.
	Save as	When the data is opened from the saved file, the currently displayed filter criteria are saved under a different name.
	Delete	Select a saved file and delete.
	Print	Print a current report.
	Update	Update a currently opened report to the latest one.
	Close	Close the report window.
Display	Caution	<Cannot be selected.>
	Description	<Cannot be selected.>
Help	Help topics	Shows help topics. (Show in English)
	Contact	Shows contact information
	Version info	Displays application name and version information.

■ Button

Button	Description
Apply	Extract data in a report based on set filter criteria.
Reset	Reset the set filter criteria and return to the default value.

3.2.2 Filter Criteria

The table below is the filter display criteria set on the [Report].



No	Filter criteria	Description
1	Type	Specify a report type. For the report type, refer to 3.1.1.
2	Data Range	Specify a date period.
3	Equipment	Specify a machine.
4	Line Name	Specify a line name.
5	Zone/Area name	Specify a zone/area name.
6	Zone name	Specify a zone name.
7	Lane	Specify a lane number.
8	Stage Number	Specify a stage number.
9	Part Number	Specify a part number.
10	Vendor No	Specify a vendor.
11	Lot No	Specify a lot number.
12	UserData1	Specify UserData1.
13	Operator Name	Specify an operator name.
14	Material Barcode	Specify material barcode.
15	MSD Level	Specify a MSD level.
16	Cart ID	Specify a cart ID.
17	Work Order	Specify a work order.
18	MCID	Specify MCID.
19	Model ID	Specify a model ID.
20	Serial No	Specify a serial number.
21	PCB Barcode	Specify a PCB barcode.
22	Nozzle Block Lit Serial Number	Specify a nozzle block kit serial number.
23	Equipment File name	Specify a machine file name.
24	PCB keep reason	Specify a PCB keeping reason.
25	AOI machine	Specify an AOI machine.
26	Feeder Barcode	Specify feeder barcode.
27	Location	Specify a location.
28	Status	Specify status.

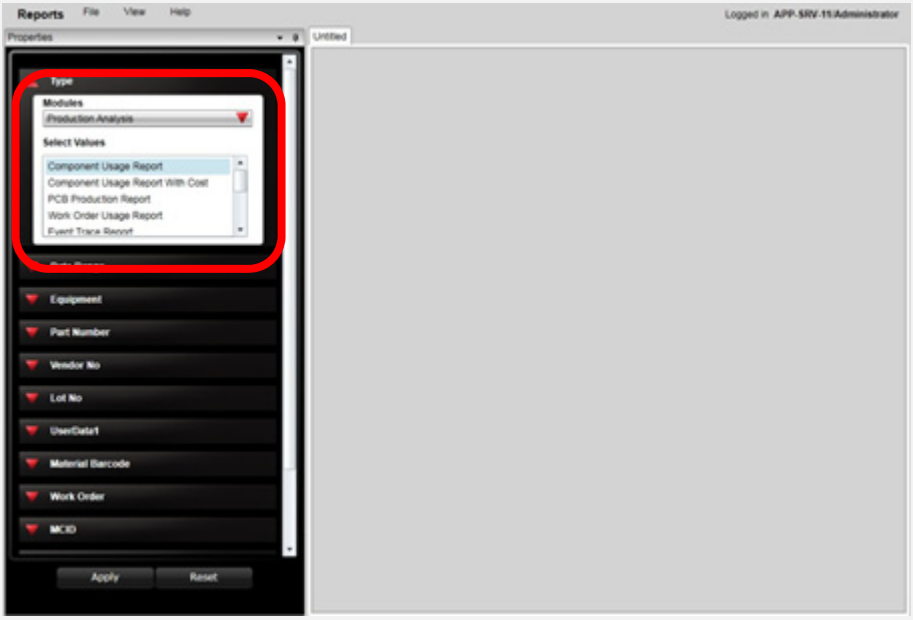
No	Filter criteria	**Description**
29	Quantity	Specify default quantity, predicted quantity, reserved quantity, default quantity (Splice), total quantity together with operator. (Multiple selections are allowed)
30	Setup ID	Specify a setup ID.
31	Material consumed reason	Specify a material consumption reason.
32	Machine group	Specify a machine group.
33	Error Code	Specify an error code.
34	Event	Specify an event.
35	Machine stop reason	Specify a machine stop reason.
36	Status	Specify a status.
37	Group By	Specify an item to be counted by group.
38	Export	Set whether data is exported or not, timing of export and export options.

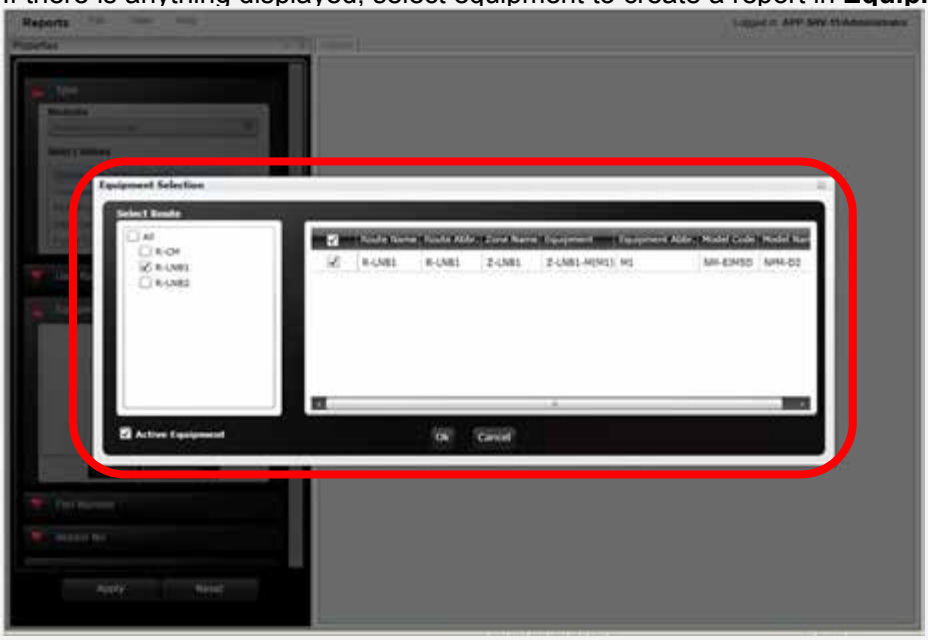
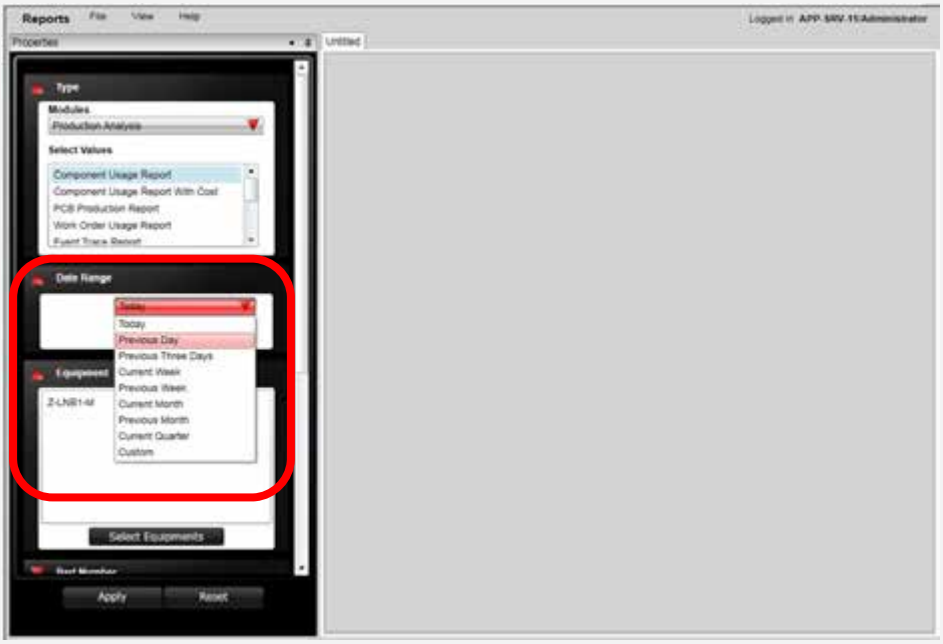
Display items very depending on a report.

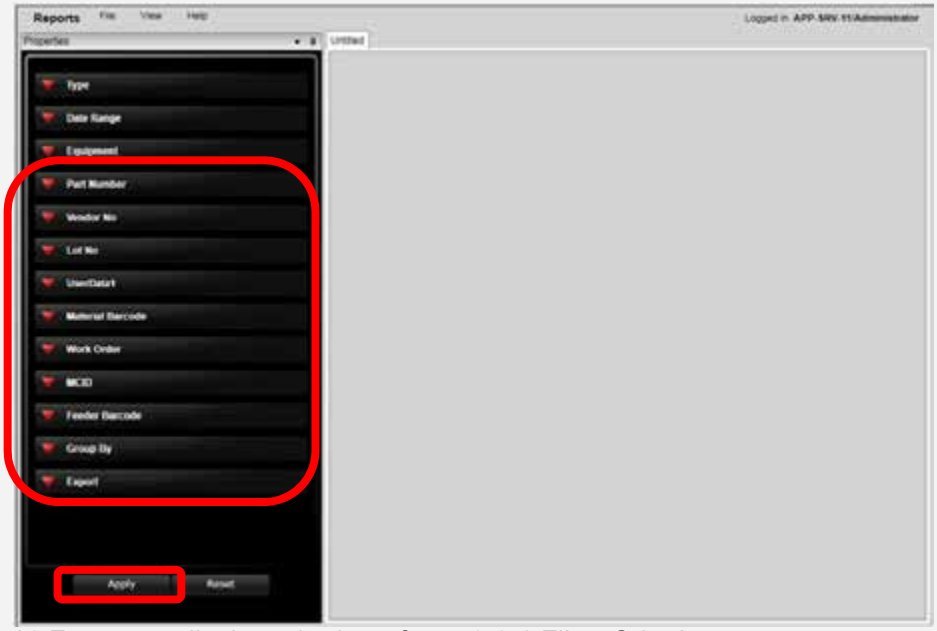
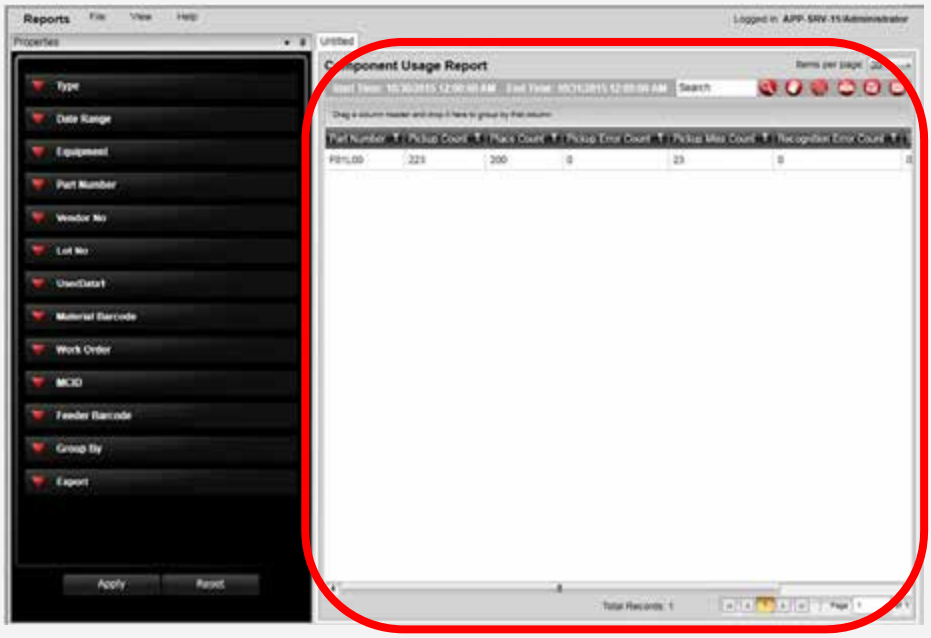
3.3 Operation

3.3.1 Operation Procedure

You can create a report by selecting ‘Report’ from the PanaCIM Web application.

Step	Operation
1	Launch PanaCIM Web application.
2	Select **Report** from **Chart and Report menu**. 
3	The following items are selected and configured in **Type**. ■ **Modules**：Select a module of a report to be displayed. (For a list of reports, refer to 3.1.1 List of Reports.) ■ **Select Values**： Select a report that you want to display from the list. 

Step	Operation
4	If there is anything displayed, select equipment to create a report in **Equipment**. 
5	If there is anything displayed, set data and time of a report in **Date Range**. 

Step	Operation
6	If there are display criteria (e.g. part number, vendor, lot number, etc.) *$_1$ of a report, configure the settings and upon completion of criteria settings, click **apply**.  *1 For report display criteria, refer to 3.2.1 Filter Criteria.
7	A report is created and displayed in the window.  • You can save the settings so far as a configuration file (template) by clicking **Save** in the file menu.

4. FUNCTION - Alarms and Exceptions

This chapter describes the display of alarms and exceptions.

4.1 Overview



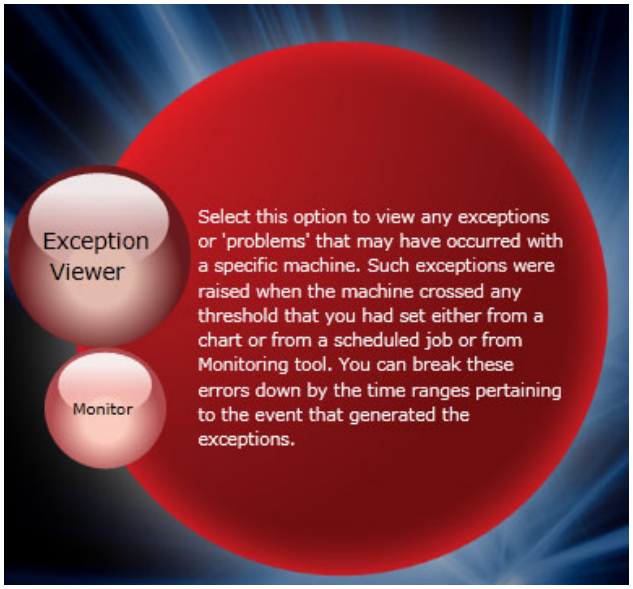
Set event log, status refresh, error log and exception log display. This chapter explains the following options.

Exception Viewer	Display exceptions or problems occurred in machine. Errors can be categorized by the time when the event occurred.
Monitor	Monitor machine's operational status and send alarms.

4.2 Settings

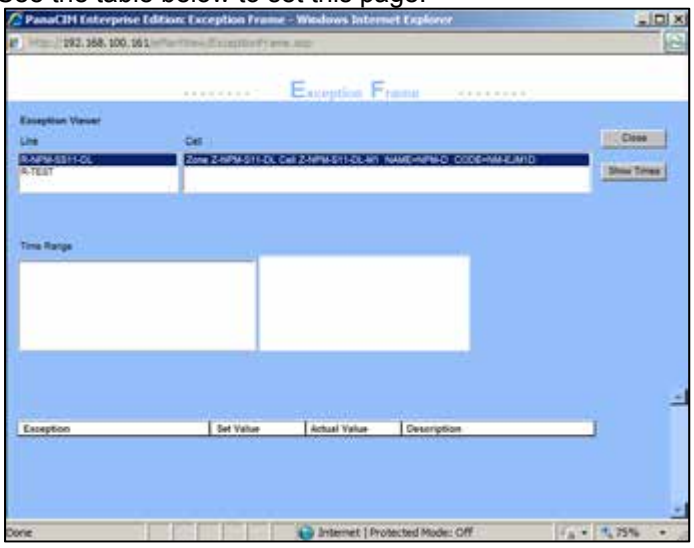
4.2.1 Exception Viewer

Exceptions occurred in machines are displayed. Exception occurs when machine's function values go beyond the control limit.



■ Exception Frame Page

See the table below to set this page.



Step	Operation
1	Select a line and machine where exception occurred.
2	Click **Show Times**. Machine's time check appears next to the image (when available).
3	Select **Date** and **Time** to display exceptions.
4	Click **Advanced**. All occurred exceptions appear at the bottom of the window.
5	Click **Printer** and print the exception log.
6	Click **Close**.

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5. FUNCTION - Administration

This chapter describes administration function.

5.1 Overview

Administration function can be used only by the users who belong to Administrators group on Windows. Windows User Management group or domain account is supported. Users in Administrators group can access to all functions, however the functions other users can access are different.



This chapter explains the following options.

Work Order Configuration	Notify work order refresh by email. See Material Control manual for more details.
User Administration	Manager users' access right.
Task Administration	Show a list to overview jobs that have been created based on the last executed time or next execution time. Jobs can be executed, disabled or deleted from this list.
Calendar Administration	Set start time for the first quarter or week or day. You can also set a desired time range.
Shift Administration	Set production shifts.
Data Import	This function is not available.

5.2 Settings

5.2.1 User Administration

Manage accessible charts, report types and routes on a per user basis.



User's access right can be set individually for active databases and archived databases.

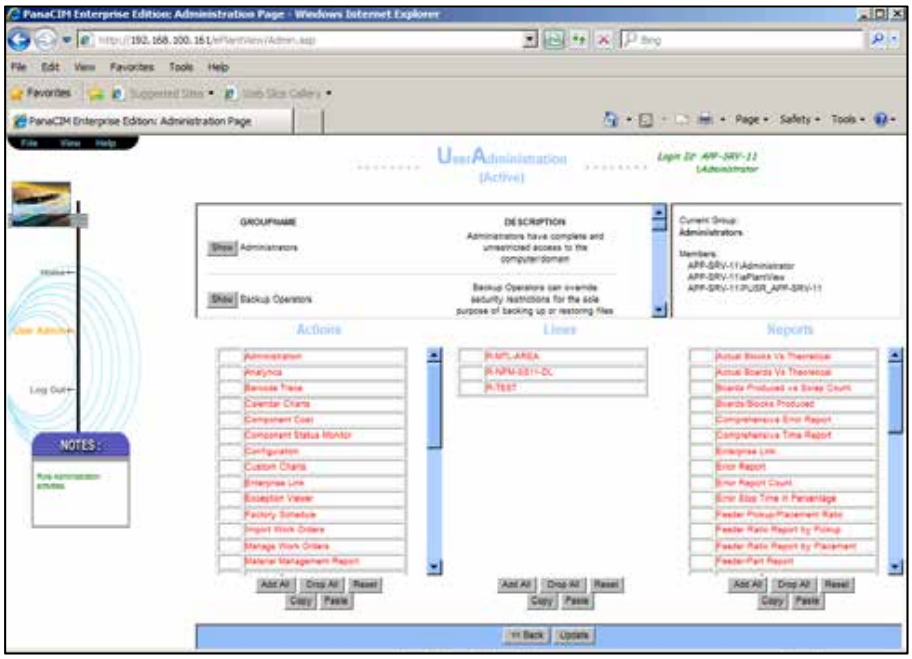
Select a database before you select **User Administration**.



- Current data: Select **Active**.
- Old data: Select **Historical**.

■ User Administration Page

Current or old data page have the same buttons and contents.



See the table below to set the page.

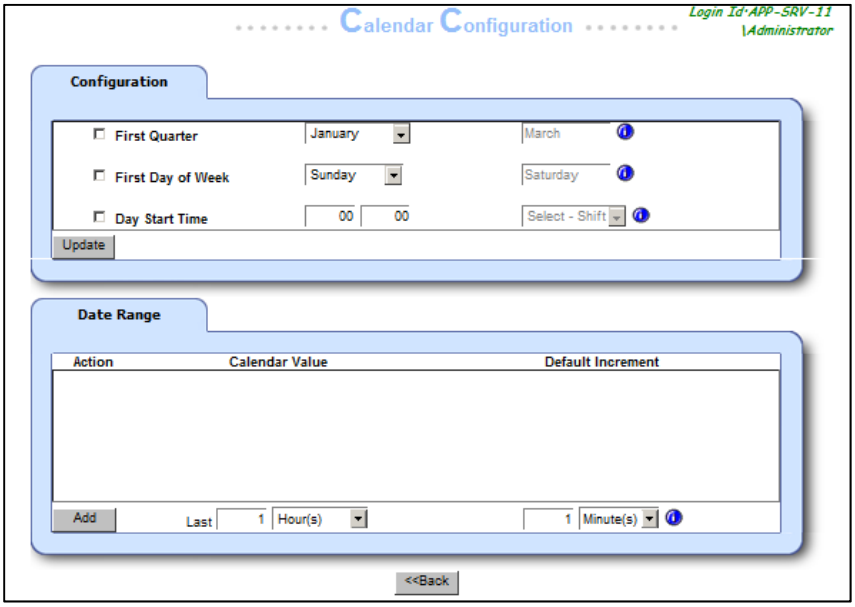
Field	Description
Action	Shows functions that the selected group can access to.
Line	Shows lines that the selected group can access to.
Report	Shows reports that the selected group can access to.
Button	**Description**
Show	Selected group members appear in the small window on the right, and their access right appears in the list below. The items that the selected group can access to are displayed in red.
Add All	Add all selected items.
Drop All	Clear the selection.
Reset	Reset the selection.
Copy	Copy selected option(s) to clipboard.
Paste	Select a group to paste the copied option and click **Paste**.
Update	Update the changes.
Back	Go back to PanaCIM Web Application top page.

5.2.2 Calendar Administration

Set a calendar to use as chart/report condition. Or you can also add an existing time range.



■ Calendar Configuration Page



The following settings are available at **Configuration** field on Calendar Configuration page.

- First month of a quarter
- First day of a week
- Start time of a day

See the table below and set the page.

Operation	Step
Change the first month of a quarter	Select First Quarter and an option from the drop-down list or enter a month. To apply the setting, press **Update**.
Set the first day of a week	Select First Day of Week and an option from the drop down list or enter a day. To apply the setting, press **Update**.
Set the start time of a day	Select Day Start Time and an option from the drop-down list or enter the time. To apply the setting, press **Update**.

DATE RANGE field on Calendar Configuration page can use the **DATE RANGE** displayed in the list when pre-defined value was selected on <Reports> Period Selection screen.



Step	Operation
1	Enter a calendar value.
2	Enter the initial time increment (default value for data display interval in graph).
3	Click **Add**.

5.2.3 Shift Administration

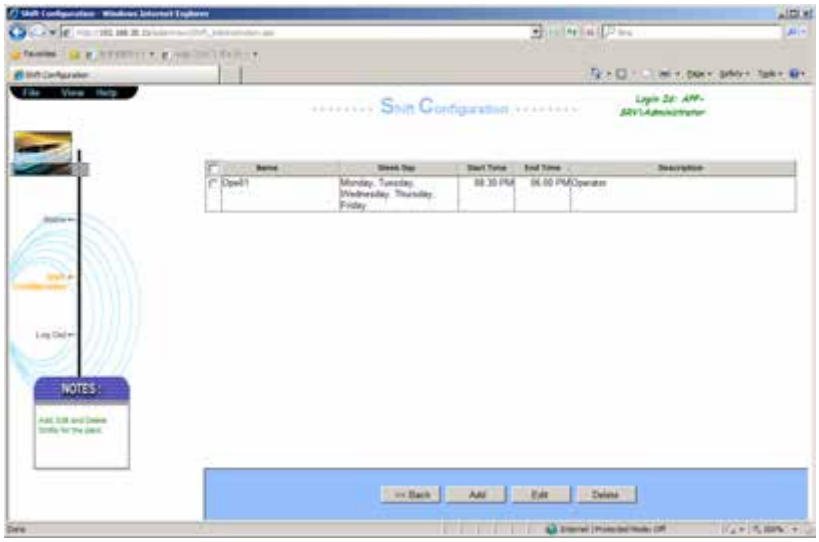
Set and manage shifts. Setting shift allows you to select shift within the data range in **Report**.





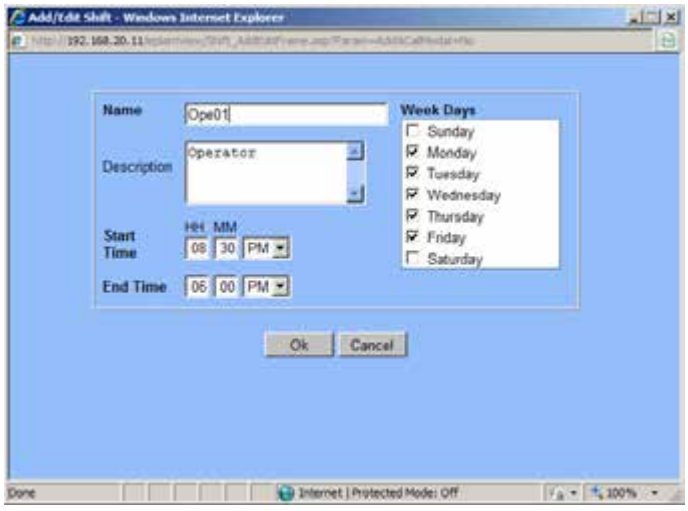
■ Shift Configuration Page

After you select **Shift Administration**, the following Shift Configuration page appears. Set a name and details for the displayed shift. Set also the date, start time and end time for shifts.



Note: The screen shows the details of a shift that is to be set on the next page in English.

● Add/Edit Shift



See the table below to add or edit shift.

Step	Operation
1	Click **Add** or **Edit** on <Shift Configuration> page. <Add/Edit Shift> dialog appears. (Select a target shift before you click **Edit**.)
2	Enter or edit name.
3	Enter or edit description.
4	Enter or edit start time.
5	Enter or edit end time.
6	Select or edit day.
7	Click **OK**.

Example) When a production shift is from 1:00AM to 12:00PM, set as follows.
Start Time: HH – 01, MM – 00, AM
End Time: HH – 12, MM – 00, PM

- **Delete Shift**

Step	Operation
1	Select a shift.
2	Click **Delete**.
3	Click **OK** on the confirmation dialog.

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